First Assignment of MedViz Project

Scientific Visualization – Master in Innovation and Research in Informatics

Objective

Medical data visualization is a key field in scientific visualization. It helps researchers and physicians interpret complex 3D data, such as CT or MRI scans, to better understand anatomy, detect patterns, and support decisions.

In this assignment, you will explore how to **transform raw medical image data into clear visual representations** using **SimpleITK**. It is a Python library built on top of ITK, widely used for medical image analysis and visualization.

You will:

- Learn how to **load, process, and visualize** 2D and 3D medical images.
- Understand the main stages of a **medical visualization pipeline**, from preprocessing to enhancement and data export.
- Practice writing **clear, reproducible notebooks** that communicate visualization results effectively.
- Gain experience in **analyzing and reviewing** visualization work critically.

By completing this first assignment of the project, you'll develop practical skills useful in data science, biomedical imaging, and scientific communication.

Structure and Deliverables

Part 1 — Group Work (in class) [2 hours]

Work in **groups of 2–3 students** to create a Google Colab notebook (include your names on it) that demonstrates the key stages of a medical visualization workflow.

Goal: Your group should produce a working, well-documented Simple ITK guide covering four stages clearly and correctly.

Sections you **need** to cover:

1. **Introduction to SimpleITK**

Overview, installation, and its role in medical visualization.

2. **Loading and Displaying Images**

Load and visualize a 2D or 3D medical dataset (SimpleITK sample data or a small open dataset). Display slices or overlays.

3. **Basic Image Manipulation**

Show resizing, resampling, cropping, and intensity transformation.

4. **Image Registration**

Implement a basic registration example (e.g., rigid or translation). Visualize before / after alignment.

Submission Deadline ***(individual)****: November 6th 23:59.*

Part 2 — Individual Work (homework) [2 hours]

Each student needs to extend the group notebook with **their own version of two new sections**. These are **personal** and should show individual insight or creativity.

Sections you **need** to cover:

5. **Visualization Enhancement**

Apply and compare visualization methods (color maps, overlays, annotations, segmentation masks, or 3D rendering). Explain your choices and what they reveal about the data.

6. **Format Transformation**

Convert the processed data to `.raw` format. Follow the naming scheme:

```
NAME_DIMX_DIMY_DIMZ.raw.
```

Include a short discussion of file formats in medical visualization (advantages / disadvantages).

Submission Deadline ***(individual)****: November 13th 23:59.*

Part 3 — Individual Peer Review (homework) [1 hour]

You will review the **document** of an assigned classmate. You will need to provide:

- **3–4 visible comments** as comments in the notebook, markdown text or code (e.g. > 💬 `Comment by [Your Name]: Nice colormap choice!`)
- A **short feedback paragraph (≈ 150 words) in a new section** covering:
    - Clarity of explanation
    - Visual and code quality
    - Suggestions for improvement
- A **grade (0–10)**

Submission Deadline ***(individual)****: November 20th 23:59.*

Evaluation Criteria

This warm-up will cover the **10% of the grade of the project**. It will be evaluated considerin the following criteria:

Criterion	Description	Weight
Global group work	Group notebook covers stages 1–4	25%
Individual extension depth	Creativity and correctness in sections 5–6	25%
Peer review quality	Constructive and specific feedback	15%
Clarity & organization	Code and text are readable and well structured. It is technically precise and concise.	25%
Visual & communication quality	Effective use of visuals and explanations	10%

Notes

- Use SimpleITK sample data (e.g., `sitk.GetSampleData('BrainProtonDensitySlice')`) or small open volumes (e.g. in racó).
- Comment your code generously — clarity matters more than complexity.
- Focus on readability and insightful visuals rather than long reports.
- Creativity is welcome!